

CHAPTER 9 : ENVIRONMENTAL MANAGEMENT

Topic:-

Visit a nearby hospital and collect information about how waste is managed.

Introduction:-

Hospitals, nursing homes, clinics, and pathological laboratories are vital institutions dedicated to diagnosing diseases, treating illnesses, and saving human lives. However, in the course of these noble healing operations, healthcare facilities generate a highly dangerous and specialized category of refuse known as **Biomedical Waste (BMW)**.

Biomedical waste includes anatomical human tissues, blood-soaked cotton, used surgical bandages, discarded intravenous catheters, hypodermic syringes, sharp scalpel blades, diagnostic reagents, and expired medicines. If this toxic waste is carelessly dumped along with ordinary municipal garbage, it can spread fatal infectious illnesses (such as Hepatitis B, Hepatitis C, and HIV), contaminate drinking water aquifers, and cause severe puncture injuries to sanitation workers and community animals.

To prevent environmental and public health catastrophes, the Ministry of Environment, Forest and Climate Change enacted the stringent **Bio-medical Waste Management Rules**. For this project, our student team conducted an educational field study visit to a 100-bed government sub-district civil hospital to examine how biomedical waste is segregated, handled, treated, and scientifically disposed of.

❖ Fundamental Principles of Biomedical Waste Management:-

- Mandatory segregation at source using standard color-coded, puncture-proof bins and bags.
- Biohazard symbol labeling on every collection container to warn healthcare staff and handlers.
- Specialized treatment methods: High-temperature Incineration, Autoclaving, and Shredding.

Aim & Hospital Visit Setup:-

Aim: To visit a certified local hospital, interview the infection control officer, inspect waste segregation protocols, and document scientific methods of biomedical waste treatment and disposal.

❖ Objectives of the Field Study:-

- To inspect the color-coded waste segregation system implemented across patient wards, OT, and labs.
- To observe the handling of sharp items (needles, scalpels) using needle-burners and puncture-proof boxes.
- To trace the transport of segregated waste from hospital wards to the Central Storage Facility.
- To understand the operation of Common Bio-medical Waste Treatment Facilities (CBWTF).

❖ Investigation Materials & Personal Safety Gear:-

- Official school authorization letter and questionnaire logbook for the Hospital Superintendent.
- Personal protective equipment: N95 protective face masks, disposable latex gloves, and lab coats.
- Camera for photographic documentation of color-coded bins in non-restricted utility corridors.



Field Visit Observations & Protocol:-

We visited the District Sub-Hospital on Friday, 22nd September. Dr. Sunita Kulkarni, the Infection Control Officer, guided us through the hospital's waste stream management:

❖ Key Operational Steps Observed in Hospital Wards:-

- **Step 1 (Segregation at Source):** Waste is categorized into dedicated color bins placed right at the bedside, nursing station, operation theatre, and casualty ward. Segregation never happens later.
- **Step 2 (Immediate Sharp Mutilation):** Disposable needles are immediately burned and severed at the tip using electrical needle destroyers at nursing stations to prevent accidental needle-stick injuries.
- **Step 3 (Bar-Coding & Bag Sealing):** When color-coded bags reach three-quarters full, they are tied tightly with zip tags and labeled with bar-coded stickers identifying the generating ward and timestamp.
- **Step 4 (Internal Transport via Dedicated Carts):** Hospital housekeeping staff use covered, wheel-mounted carts reserved strictly for biohazard bags, following designated service elevators.
- **Step 5 (Central Storage & 48-Hour Collection):** Bags are stored in a secured, tiled Central Biomedical Waste room. A GPS-tracked authorized CBWTF vehicle collects the waste every 24 to 48 hours.

Observations & Waste Categorization Table:-

Based on our inspection, we compiled the official color-coding segregation chart enforced under Bio-medical Waste Management Rules:

Color Bin	Type of Hospital Waste	Treatment Method	Final Fate / Disposal
Yellow Bin (Non-chlorinated)	Human tissues, organs, placenta, soiled cotton, bandages, expired drugs	High-temperature Incineration (>1050°C)	Controlled ash burial in secure landfill
Red Bin (Non-chlorinated)	Contaminated plastic waste, catheters, urine bags, syringes without needles	Autoclaving (steam sterilization) + Shredding	Recycled by authorized plastic recyclers
White Translucent (Puncture-proof)	Sharps, needles, scalpels, surgical blades, contaminated broken glass	Autoclaving / Dry Heat + Sharp shredding	Encapsulation in concrete pit
Blue Cardboard / Plastic Box	Medicine vials, ampoules, broken antibiotic bottles, metallic implants	Sodium hypochlorite soaking / Autoclaving	Glass recycling units
Black / Green Bin	General non-hazardous waste (paper, fruit peels, food leftovers)	Standard Municipal Handling	Composting / Municipal landfill

Yellow and Red categories comprise approximately 85% of the total biomedical waste generated daily.

Scientific Analysis & Treatment Technologies:-

Scientific evaluation of biomedical waste processing highlights why thermal and sterilization technologies are mandatory:

❖ *Core Scientific Mechanisms Used in Waste Treatment:-*

- **Incineration (Yellow Waste):** Dual-chamber incinerators operate with a primary chamber at 800°C to 850°C to gasify organic pathogens and a secondary combustion chamber at 1050°C ($\pm 50^\circ\text{C}$) with 2 seconds gas residence time to destroy deadly dioxins, furans, and highly resistant viral capsids completely.
- **Autoclaving (Red Plastic Waste):** Uses saturated steam under high pressure (121°C at 15 psi for 45 minutes). Steam penetration denatures all bacterial spores (e.g., *Bacillus stearothermophilus*), rendering plastics completely sterile for clean recycling.
- **Chemical Disinfection:** Liquid laboratory wastes (blood, pleural fluid) are pre-treated on-site with 1% to 2% sodium hypochlorite (bleach) solution for at least 30 minutes before discharge into effluent treatment plants (ETP).

Public Health Hazards & Key Takeaways:-

Improper disposal of biomedical waste poses severe multi-faceted threats to human communities and wildlife:

❖ Dangers of Improper Biomedical Waste Disposal:-

- **Transmission of Blood-Borne Pathogens:** Needlestick injuries from discarded hypodermic syringes transmit lethal viruses (HIV, Hepatitis B, Hepatitis C) to ragpickers and sanitation workers.
- **Toxic Fumes from Open Burning:** Burning medical plastics (PVC tubes) in open dumps emits carcinogenic dioxins, furans, and hydrochloric acid gas into neighborhood air.
- **Antimicrobial Resistance (AMR):** Discarding active antibiotics into general drains breeds "superbugs"—bacteria resistant to all known medicines—in community water bodies.

❖ Household Parallels for Daily Citizen Safety:-

- Wrap used bandages, sanitary napkins, and diapers securely in newspaper and mark with a red dot.
- Never discard expired household medicines into the kitchen sink; return them to pharmacy drop-offs.

What We Learned, Conclusion & References:-

❖ What We Learned From This Project:-

- Learned the scientific categorization, color-coding, and disposal pathways of biomedical waste.
- Observed practical hospital safety procedures and the crucial role of infection control personnel.
- Realized that proper waste segregation at source protects both sanitation workers and the environment.

❖ Project Conclusion:-

Our visit to the local civil hospital provided eye-opening practical exposure to biomedical waste management. We observed that hospital safety depends on rigorous color-coded segregation right at the point of waste generation. When hospitals adhere strictly to Bio-medical Waste Management Rules, infectious dangers are neutralized efficiently, safeguarding both public health and ecological integrity.

❖ Bibliography & References:-

- Maharashtra State Bureau of Textbook Production (Balbharati) – Standard 9 Science & Technology (Chapter 9: Environmental Management).
- Ministry of Environment, Forest and Climate Change (MoEFCC) – Bio-medical Waste Management Rules (2016, amended 2019).
- Central Pollution Control Board (CPCB) – Guidelines for Management of Healthcare Waste in India.